- Profile pages for musicians, bands, and venues
- Classified ad listings
- Announcement listings for tryouts
- Search capabilities based on instruments played and style of music
- Messaging between potential matches

Nicky has confidence in your coding skills because you've talked excitedly about the Python you've learned in the past. As you think about the problem, you realize that those features are just the business requirements. To build a working website that can do all those things, you'll also need features that are common to most websites, such as

- Public-facing and private-only web pages
- Logins
- Ability for users to submit both text and image content
- User sign-up
- Password management
- A database for storage
- Admin pages for managing the site

A lot goes into what seems like a simple website. Asking around, you've heard that Django provides all those capabilities for you, so you've decided to learn Django to help you build RiffMates.

Django is a *framework*. Programming with a framework is different than programming with a library. In the case of a library, your code determines the order of execution. By contrast, when you're building inside a framework, execution control belongs to the framework. If you were building a house, a library would contain the pieces you needed for the house, whereas a framework is already the rough outline of the house. Working within a framework means your application designs are more constrained, but what you lose in flexibility, you gain in efficiency—there is less code for you to write.

The partially completed building that you get with Django is the structure of a website. Django calls this a *project*. A Django project is what interacts with a web server to respond when a user visits a URL. Your code lives inside the project and gets invoked by Django. Each time you want to create a new site, you create a new project, and you need to have the project in place before you can start adding pages to it. The rest of this chapter covers creating a project and writing your first web page inside it.

2.2 Creating a Django project

The Django framework includes a command-line tool that is used for creating new projects, called `django-admin` (not to be confused with the Django Admin, the web tool for managing your data). Django is published on PyPI (https://pypi.org/) and can be installed in a Python virtual environment using the command `python -m pip install django`. Once you've got the package installed, you use the command `django-admin` to create a skeleton framework for your project.